

Self-Distilled Agentic Reinforcement Learning

Zhengxi Lu^{1,2*}, Zhiyuan Yao^{1,2}, Zhuowen Han², Zi-Han Wang^{2,3}, Jinyang Wu³
Qi Gu^{2†}, Xunliang Cai², Weiming Lu¹, Jun Xiao¹, Yueting Zhuang¹, Yongliang Shen^{1†}
¹Zhejiang University ²Meituan ³Singhua University
{zhengxilu, syl}@zju.edu.cn guqi03@meituan.com

Abstract

Reinforcement learning (RL) has emerged as a central paradigm for post-training LLM agents, yet its trajectory-level reward signal provides only coarse supervision for long-horizon interaction. On-Policy Self-Distillation (OPSD) complements RL by introducing dense token-level guidance from a teacher branch augmented with privileged context. However, transferring OPSD to multi-turn agents proves problematic: compounding multi-turn instability destabilizes supervision, while skill-conditioned privileged guidance requires asymmetric treatment for negative teacher rejections may arise from imperfect skills retrieval or utilization. We introduce SDAR (Self-Distilled Agentic Reinforcement Learning), which treats OPSD as a gated auxiliary objective while keeping RL as the primary optimization backbone. SDAR maps detached token-level signals into a sigmoid gate, strengthening distillation on teacher-endorsed positive-gap tokens and softly attenuating negative teacher rejections. Across the Qwen2.5 and Qwen3 families on ALFWorld, WebShop, and Search-QA, SDAR substantially improves over GRPO (+9.4% on ALFWorld, +7.0% on Search-QA, +10.2% on WebShop-Acc), avoids the instability of naive GRPO+OPSD, and consistently outperforms hybrid RL-OPSD baselines across model scales. Code available: <https://github.com/ZJU-REAL/SDAR>.

Figure 1: (a) Comparison between GRPO+OPSD and SDAR; (b) Overall Performance.

*Work done during internship at Meituan.

†Corresponding author

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1 Introduction

Agentic post-training has become a central challenge for Large Language Models (LLMs) (Guo et al., 2025; Team et al., 2025; Yang et al., 2025; Comanici et al., 2025; Team et al., 2026b). Unlike static single-turn reasoning, multi-turn agents interact with environments over extended horizons, where each action changes future observations and each generated response becomes part of the context for subsequent decisions (Shen et al., 2023; Shi et al., 2025; Jimenez et al., 2023).

Two paradigms naturally emerge as complementary forces: Reinforcement Learning (RL) (Shao et al., 2024; Dong et al., 2025; Feng et al., 2025) provides task-level optimization grounded in environment or verifier feedback, whereas On-Policy Distillation (OPD) (Ye et al., 2026; Yang et al., 2026b; Team et al., 2026a; GLM-5-Team et al., 2026) and On-Policy Self-Distillation (OPSD) (Zhao et al., 2026; He et al., 2026; Zhang et al., 2026) provide dense token-level guidance from a teacher branch. Yet, OPSD does not transfer cleanly to multi-turn agent training. We attribute this to two observations: [1] Multi-turn OPSD Instability and [2] Asymmetric Trust in Privileged Guidance.

[Observation-1] Multi-turn OPSD Instability Once the student agent inevitably drifts

0 50 100 150

Training Steps

0.00

0.05

0.10

0.15

0.20

0.25

0.30

0.35 KL Loss

GRPO+OPSD

GRPO

RLSD

0 5 10 15 20

Turn Step

1.0

1.5

2.0

2.5

3.0 KL Divergence

Multi-turn OPSD

Training Steps

0.1

0.2

0.3 Success Rate

Multi-turn OPSD

Figure 2: Left: Multi-turn OPSD Instability, with performance

and KL reported. Right: RLSD-Style Instability, with KL loss.

from the teacher-supported trajectory, the once-helpful token-level supervision becomes increasingly unreliable. This compounding error leads to surging per-turn KL divergence and catastrophic degradation in task performance, as shown in Figure 2 (Left). TCO (Wang et al., 2026b) attempts to address this through curriculum learning, but relies on rigid temporal schedules or trajectory-depth thresholds.

[Observation-2] Asymmetric Trust in Privileged Guidance. In OPSD, the teacher branch is not an independently stronger model, but the same policy augmented with privileged training-only context, such as retrieved skills. This makes its token-level guidance inherently asymmetric. For a student-sampled token y_t , if the privileged teacher assigns a higher probability than the student, the retrieved skill provides an endorsement signal: it supports an on-policy behavior that the student can already generate but has not fully internalized. Such positive guidance is particularly suitable for distillation.

In contrast, if the privileged teacher assigns a lower probability to the sampled token, the signal should be interpreted more cautiously. A negative gap may indicate that the token should indeed be suppressed, but in skill-conditioned OPSD it may also arise from the instability of privileged context: (1) Skill Quality. Retrieved skills may be irrelevant, incomplete, or redundant. (2) Skill Utilization. The teacher may fail to ground even relevant skills into reliable token-level preferences (Chen et al., 2019). (3) Multi-turn Drift. As trajectories unfold, the teacher-student gap tends to widen across turns (Figure 3, Middle), amplifying early mismatches over successive decisions (Ross et al., 2011). Our preliminary study on Qwen2.5-3B-Instruct shows that negative-gap tokens exceed 50% of all tokens (Figure 3), making this issue pervasive. This motivates an asymmetric treatment of privileged guidance: trust positive teacher endorsements more strongly, while applying negative teacher rejections more conservatively.

A stark realization emerges: for multi-turn agents, RL could reign as the primary optimization backbone, while OPSD is relegated to a carefully controlled auxiliary role. But how should this auxiliary role be controlled? RLSD (Yang et al., 2026a) directly uses self-divergence to re-weight token-level RL advantages, but can substantially amplify updates especially early in training when teacher-student mismatch is large (see Figure 2, Right).

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-1 -0.5 0 0.5 1

Teacher-Student Gap

0

50000

100000

150000

200000

250000 Token Count

Mean=-0.1168

1 11 21 31 41

Turn Step Interval

0.11

0.12

0.13

Average Gaps

5% 30% 55% 80%

Relative Position in Turn

0.18

0.16

0.14

0.12

0.10

0.08

0.06

0.04

Average Gaps

Figure 3: Teacher-Student Gap Analysis. Left: Token count distribution partitioned by Teacher-Student gap value. Middle: Average teacher-student gap indexed by multi-turn step. Right: Average teacher-student gap indexed by relative position within a single turn.

We take a different path: the OPSD loss is treated as a direct, auxiliary optimization objective, leaving the verifier-driven RL policy loss untouched and thereby strictly preserving the semantics and unbiasedness of the RL advantage. To overcome instability of multi-turn OPSD and privileged guidance, distillation is not performed uniformly on every token. Instead, tokens are selectively distilled via an adaptive, smooth gating mechanism rather

than a hand-crafted, rigid schedule (such as Skill-SD (Wang et al., 2026a) and HDPO (Ding, 2026)). Inspired by TIP (Xu et al., 2026), we use token-level signals (such as student entropy or teacher-student divergence) to control the gate’s activation. The core philosophy is simple: let each token decide the intensity of its own supervision. This yields a dynamic, self-paced curriculum operating at the finest possible granularity: the individual token level. We validated our method across the Qwen2.5 and Qwen3 model families on three diverse benchmarks for llm-based agents: ALFWorld (Shridhar et al., 2020), WebShop (Yao et al., 2022), and Search-QA (Jin et al., 2025). SDAR achieves substantial improvements over GRPO (+9.4% on ALFWorld, +7.0% on Search-QA, and +10.2% on WebShop-Acc for 7B), entirely avoids the catastrophic instability of naive GRPO+OPSD, and consistently outperforms RL-OPSD hybrid methods such as Skill-SD and RLSD across all three model scales (Qwen3-1.7B included). Furthermore, robustness analysis shows that SDAR degrades gracefully with retrieval quality: even random retrieval outperforms the GRPO baseline, as our gating design filters out noise from low-quality skills and distills beneficial signals only.

2 Method

2.1 Problem Setup

We consider a multi-turn agent that interacts with an environment over a finite horizon. Given an initial prompt or task description x , at turn k the agent receives an observation o_k , generates a response a_k , and the environment returns the next observation o_{k+1} . Each response a_k may contain both intermediate reasoning tokens and executable action tokens. For notational simplicity, we flatten all valid response tokens in one trajectory into a single token sequence

$$y = (y_1, \dots, y_T) \sim \pi_\theta(\cdot | x),$$

where π_θ denotes the student policy and T is the total number of valid response tokens. At token position t , we denote the self-student context by

$$s_t = (x, y_{<t}),$$

and the self-teacher context by

$$s^+ =$$

$$t = (x, c^+, y_{<t}),$$

where c^+ denotes privileged training-only context available only to the teacher branch, such as reference answers, skills (ours), or other auxiliary information not accessible at test time.

3

Environment

Turn 1

Turn N

r_1

r_2

r_G

τ_1

τ_2

τ_G

A_1

A_2

AG

Skills

Retrieval

Self-Teacher

Gaps

Rollout

OPSD Agent Loop GRPO

Self-Student

Given the Current Situation , it seems

the side table 1 is empty

-0.25 0.06 -0.07 0.02 0 0.46 -0.43

0.20 -1.7 0.00 0.0 -0.03 0.28

Token-Level Gating

Token-Level Advantages

Tokens λ^* OPSD Loss + GRPO Loss

Task-Outcome Signal

Agent

Env

Action

Observation

Traj Reward

Figure 4: Illustrations of SDAR framework, which trains multi-turn agents using token-level OPSD loss and verifier-driven RL loss.

Skills Retrieval We retrieve task-relevant skills—compact, structured demonstrations that encode domain-specific knowledge such as sub-goal decompositions or action templates.

We implement four retrieval strategies of varying quality to evaluate the robustness of our framework to the fidelity of the retrieved context: (1) UCB Retrieval, (2) Keyword Matching (KM), (3) Full Retrieval, and (4) Random Retrieval.

Skill retrieval is cast as a multi-armed bandit problem over the skill library $E = \{e_1, \dots, e_M\}$.

For each incoming task, UCB Retrieval selects the single highest-scoring skill file according

to the Upper Confidence Bound (UCB) criterion:

$$\text{score}(e) = \bar{r}(e) + c$$

s

$\ln N$ ucb

$n(e)$, (1)

where $\bar{r}(e)$ is the running mean reward obtained when skill e was previously supplied as context, Nucb is the total number of retrieval queries issued for the same task type, $n(e)$ is the number of times e has been selected, and c controls the exploration-exploitation trade-off.

Keyword Matching bypasses the bandit formulation and instead identifies the task scenario by matching keywords in the task description against predefined category labels, directly retrieving the skill file associated with the matched category.

2.2 Optimization Goals

Our method is designed as an auxiliary objective on top of a standard policy optimization

GRPO loss. The overall training objective is

$$L(\theta) = L_{\text{GRPO}}(\theta) + \lambda \text{SDAR} \cdot L_{\text{SDAR}}(\theta),$$

where L_{GRPO} is the original policy loss and L_{SDAR} is our on-policy self-distillation objective.

Let $m_t \in \{0, 1\}$ be the response mask indicating whether token t is valid. We define masked token averaging as

$$\text{Agg}(z_{1:T}) = \sum_{t=1}^T m_t z_t$$

$$\sum_{t=1}^T m_t$$

$$\sum_{t=1}^T m_t$$

$$\sum_{t=1}^T m_t$$

.

RL Optimization For each input x , GRPO samples a group of responses

$\{y(i)\}_{i=1}^G$

$$i=1 \sim \pi_{\theta}(\cdot | x),$$

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and computes a sequence-level advantage $A(i)$ from environment rewards. Using a reference policy π_{ref} , the GRPO objective can be written as

$$L_{\text{GRPO}}(\theta) = - \frac{1}{G} \sum_{i=1}^G \text{Agg} \left[\min \left[\begin{aligned} & r(i) \\ & t A(i), \text{clip}(r(i), 1-\epsilon, 1+\epsilon) A(i) \end{aligned} \right] \right. \\ \left. + \beta \cdot \frac{1}{G} \sum_{i=1}^G \text{Agg} \left[\text{DKL}(\pi_{\theta}(\cdot | s(i), t)) \pi_{\text{ref}}(\cdot | s(i), t) \right] \right], \quad (2)$$

G

G

$\sum$

$i=1$

Agg

■

min

■

$r(i)$

$t A(i), \text{clip}(r(i), 1-\epsilon, 1+\epsilon) A(i)$

$t, 1-\epsilon, 1+\epsilon) A(i)$

■

$+\beta \cdot \frac{1}{G}$

G

G

$\sum$

$i=1$

Agg

■

$\text{DKL}(\pi_{\theta}(\cdot | s(i), t))$

$\pi_{\text{ref}}(\cdot | s(i), t)$

$t)$

■

, (2)

where $r(i)$

$t = \pi_{\theta}(y(i), s(i))$

$t | s(i)$

$t) / \pi_{\theta_{\text{old}}}(y(i), s(i))$

$t | s(i)$

$t)$ is the importance sampling ratio.

OPSD Optimization At a fixed token position t , the teacher and student induce conditional token distributions $\pi_T(\cdot | s + t)$ and $\pi_{\theta}(\cdot | s + t)$, respectively. The per-token reverse KL

divergence is defined as:

$$D(t)$$

$$\text{RKL} = D_{\text{KL}}$$

■

$$\pi_{\theta}(\cdot | s + t)$$

■

$$\pi_T(\cdot | s + t)$$

$$t)$$

■

$$= \sum_{v \in V} \pi_{\theta}(v|s, t) \log \pi_{\theta}(v|s, t) - \log \pi_{\theta}(y_t|s, t)$$

To efficiently derive an importance signal without computing the expensive full-vocabulary summation, we take a single-sample estimate on the student-sampled token $y_t \sim \pi_{\theta}(\cdot | s, t)$. The negation of this estimate directly yields the Teacher-Student log-probability gap Δt :

$$\Delta t = -\log \pi_{\theta}(y_t | s, t) + \log \pi_{\theta}(y_t | s, t)$$

2.3 Token-Level Gating

The key idea is to convert privileged teacher guidance into a token-level trust weight, while keeping the verifier-driven RL objective unchanged. We introduce a token-level gate $g_t \in [0, 1]$ that modulates the OPSD signal on each student-sampled token, and apply it to a sampled-token surrogate so that different gating strategies share the same optimization.

Let

$$\Delta t = \text{sg}$$

$$\log \pi_{\theta}(y_t | s, t) - \log \pi_{\theta}(y_t | s, t)$$

denote the detached Teacher-Student log-probability gap on the student-sampled token, and

$$h_t = \sum_{v \in V} \pi_{\theta}(v|s, t) \log \pi_{\theta}(v|s, t)$$

denote the student entropy at position t . We compose each raw score with the logistic sigmoid σ so that every gate is smooth, differentiable, and naturally bounded in $(0, 1)$. The sharpness parameter $\beta > 0$ controls the transition between conservative attenuation and strong activation.

We instantiate three complementary gating strategies:

1. Entropy gating: $g_t = \sigma(\beta h_t)$, targeting high-entropy positions where the student is most uncertain.

2. Gap gating: $g_t = \sigma(\beta \Delta t)$, assigning larger weights to positive-gap tokens endorsed by the privileged teacher while attenuating negative-gap tokens.

3. Soft-OR gating: $g_t = \sigma(\beta[1 - (1 - h_t)(1 - \Delta t)])$, combining student uncertainty and teacher-student gap as an alternative gating strategy.

In all cases, the gate is detached via $\text{sg}(\cdot)$, so gradients flow exclusively through the student log-probability. The token-level loss is

■ SDAR

$$L_{\text{SDAR}} = \sum_t g_t \cdot (\log \pi_{\theta}(y_t | s, t) - \log \pi_{\theta}(y_t | s, t))$$

With gap gating, the sigmoid gate implements asymmetric token-level modulation: positive-gap tokens receive stronger auxiliary distillation, while negative-gap tokens are softly attenuated. We also provide theoretical analysis of our design in Appendix A.

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Table 1: Performance on ALFWorld, Search-QA and WebShop tasks. We report the success rate (%) on ALFWorld, accuracy (%) on Search-QA, and Score/Acc (%) on WebShop (128 tasks). * means validation with skills. Best and second-best are highlighted.

ALFWorld Search-QA WebShop

Method Pick Look CleanHeat Cool Pick2 Avg NQ Triv Pop Hotp 2Wk MuS Bam Avg Score Acc

Qwen2.5-3B-Instruct

Vanilla 44.4 11.1 6.2 15.4 28.6 12.5 21.9 24.6 48.1 31.0 26.3 25.3 7.2 59.7 31.7 6.7 0.8

Skill-Prompt* 51.7 66.7 48.4 0.0 4.3 10.0 28.9 23.7 46.2 30.6 24.4 22.1 7.5 12.5 23.9 0.2 0.8

OPSD 48.8 41.7 16.7 0.0 15.8 16.7 28.1 0.1 0.1 0.1 0.0 0.0 0.0 0.0 11.3 3.1

GRPO 91.2 62.5 96.2 61.9 65.0 47.4 75.0 39.3 60.6 41.1 37.4 34.6 15.4 26.4 36.4 79.8 63.3

Skill-GRPO 88.9 71.4 58.8 70.6 40.7 29.2 60.2 43.5 58.8 43.0 36.8 32.2 11.7 12.5 34.1 77.3 60.9

Skill-GRPO* 94.3 57.1 100 66.7 73.1 57.1 80.5 44.3 59.6 44.3 39.0 36.1 14.5 14.9 36.1 76.3 66.4

GRPO+OPSD 100 82.4 85.7 75.0 70.0 60.0 81.2 44.9 61.2 45.2 40.4 38.5 16.0 66.1 44.6 77.8 66.4

Skill-SD 88.2 50.0 96.2 52.4 65.0 57.9 73.4 44.4 60.4 44.0 39.5 40.4 15.4 64.9 44.1 75.9 64.0

RLSD 87.9 75.0 90.9 75.0 73.1 68.4 79.7 41.5 58.6 42.3 40.4 40.2 16.8 66.9 43.8 84.4 66.4

SDAR 97.1 62.5 100 61.9 75.0 84.2 84.4 44.8 58.1 44.3 38.6 36.2 15.7 66.1 43.4 85.0 68.0

Qwen2.5-7B-Instruct

Vanilla 36.1 22.2 3.1 0.0 0.0 0.0 12.5 25.2 50.8 29.5 29.0 29.0 10.4 63.7 33.9 5.9 1.6

Skill-Prompt* 51.7 50.0 32.3 5.3 4.3 0.0 23.4 30.9 52.1 32.7 32.7 27.9 12.7 66.1 36.4 1.7 0.8

OPSD 50.0 60.0 22.7 21.4 17.6 9.5 32.8 8.8 8.6 17.5 2.5 4.2 0.5 1.2 6.2 4.5 2.3

GRPO 91.2 87.5 96.2 81.0 65.0 57.9 81.2 45.1 63.7 44.0 43.6 43.2 16.8 37.6 42.0 80.9 72.6

Skill-GRPO 88.5 66.7 65.2 61.1 57.7 73.1 69.5 45.2 63.7 45.7 43.1 43.3 19.6 21.4 40.3 80.4 71.9

Skill-GRPO* 100 83.3 96.4 83.3 75.0 78.9 88.3 44.8 63.0 45.1 43.7 43.7 20.5 71.4 47.5 87.0 81.2

GRPO+OPSD 91.4 61.5 100 87.5 76.5 52.2 80.4 47.3 64.5 46.9 43.8 39.3 18.0 69.4 47.0 86.8 76.5

Skill-SD 93.9 93.890.9 100 69.2 68.4 85.1 47.1 64.5 47.8 44.2 42.1 20.2 69.0 47.8 86.1 76.5
 RLSD 100 87.5 92.3 58.8 80.065.2 82.0 46.8 63.0 44.4 45.5 48.9 21.5 73.0 49.0 87.4 77.3
 SDAR 94.7 75.0 100 86.7 68.2 78.9 85.9 46.3 63.5 48.243.8 48.4 19.6 73.0 49.0 89.4 82.8
 Qwen3-1.7B-Instruct
 Vanilla 25.0 22.2 3.1 0.0 21.4 4.2 12.5 29.4 46.9 37.0 23.5 19.6 6.4 10.5 24.8 46.5 4.7
 Skill-Prompt*10.3 50.0 16.1 0.0 0.0 5.0 9.4 29.4 46.5 36.2 22.9 20.8 4.3 10.1 24.3 23.0 2.3
 OPSD 26.3 33.3 9.1 0.0 4.5 5.3 14.1 4.2 8.3 4.6 6.6 15.3 0.7 1.2 5.8 47.4 9.3
 GRPO 71.1 41.7 36.4 40.0 31.8 31.6 46.1 40.0 58.943.5 35.4 30.3 12.0 65.7 40.8 67.3 38.3
 Skill-GRPO 27.6 54.522.7 27.3 0.0 19.2 21.1 39.2 58.6 43.9 35.2 28.2 11.5 66.1 40.4 73.4 46.1
 Skill-GRPO*31.4 42.9 51.9 8.3 11.5 7.1 28.1 38.0 58.4 43.9 36.3 29.0 12.5 66.940.7 80.4 50.0
 GRPO+OPSD38.2 50.0 30.8 28.6 30.0 21.1 32.0 40.7 58.945.0 37.0 34.6 13.365.7 42.270.7 38.3
 Skill-SD 52.9 37.5 69.2 42.9 60.0 36.8 52.3 39.1 57.5 45.434.8 34.1 10.7 64.1 40.8 81.8 53.9
 RLSD 50.0 37.5 61.5 19.0 50.0 21.1 42.2 38.6 57.3 43.0 34.5 34.1 11.5 65.3 40.6 74.0 50.8
 SDAR 73.525.0 76.933.3 40.0 36.8 53.939.7 58.9 45.3 35.9 35.5 12.6 65.3 41.9 76.8 58.6
 3 Experiment

BenchmarksWe evaluate our methods on ALFWorld (Shridhar et al., 2020), Search-based QA (Jin et al., 2025), and Webshop (Yao et al., 2022). ALFWorld is a text-based game aligned with the ALFRED embodied AI benchmark, including 3,827 task instances across six categories of common household activities: Pick and Place (Pick), Look at Obj in Light (Look), Pick Clean then Place in Recep (Clean), Pick Heat then Place in Recep (Heat), Pick Cool then Place in Recep (Cool), and Pick Two Obj and Place (Pick2). Search-based QA contains several widely-used search-augmented QA benchmarks, including single-hop QA datasets (NQ (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), and PopQA (Mallen et al., 2023)) and multi-hop QA datasets (HotpotQA (Yang et al., 2018), 2Wiki (Ho et al., 2020), MuSiQue (Trivedi et al., 2022), and Bamboogle (Press et al., 2023)). WebShop is a complex, web-based interactive environment designed to test the LLM agents in realistic online shopping scenarios. Agents navigate a realistic web interface to find and purchase products matching user specifications. We select 128 fixed tasks in validation set, which aligns with Feng et al. (2025).

Implementation DetailsWe train the Qwen2.5-Instruct and Qwen3-Instruct series using SDAR for at 150 steps on 8 H800 GPUs. For ALFWorld, we adopt the training data split from GiGPO (Feng et al., 2025), with each batch sampling 16 tasks and 8 rollouts per prompt, and a maximum prompt length of 2,048 tokens. For Search-QA, we follow the experimental setup of Search-R1 (Jin et al., 2025), using E5 (Wang et al., 2022) as the retriever. The training data are drawn from NQ and HotpotQA, making these two benchmarks in-domain, while the remaining datasets serve as out-of-domain evaluation. Each batch samples 128 tasks with a maximum prompt length of 4,096 tokens. For Webshop, 1000 tasks are selected for

6 training, with each batch sampling 16 tasks and 8 rollouts per prompt, and a maximum prompt length of 4,096 tokens. We set the SkillBank from SkillRL (Xia et al., 2026) for all three environments. We set λ SDAR = 0.01 and β = 5.0 in our experiments.

BaselinesWe compare SDAR against three categories of methods on three base models.

(1) Training-free methods. Skill-Prompt retrieves task-relevant skills from the SkillBank via keyword matching (KM) and prepends them to the input prompt at inference time. (2) Post-training methods, such as GRPO (Shao et al., 2024), OPSD (Zhao et al., 2026) and Skill-GRPO. Skill-GRPO augments GRPO by retrieving skills via KM and injecting them into the training prompt; at test time it can run with (Skill-GRPO*) or without retrieved skills. (3) Hybrid methods, that combine RL with privileged knowledge distillation, such as GRPO+OPSD, and Skill-SD (Wang et al., 2026a), RLSD (Yang et al., 2026a). GRPO+OPSD simply adds the OPSD distillation loss as an auxiliary objective on top of GRPO training. All the algorithms of SDAR and other baselines are detailed in Appendix A.

3.1 Main Results

Overall PerformanceAs summarized in Table 1, SDAR demonstrates exceptional performance, achieving the best or second-best results across almost all settings. Compared to GRPO, it delivers substantial gains: on Qwen2.5-3B, it improves ALFWorld by +9.4% (84.4 vs. 75.0), Search-QA by +7.0%, and WebShop-Acc by +4.7%, with similarly consistent improvements on the 7B model. While standalone OPSD collapses catastrophically (near-zero on Search-QA) and a naive GRPO+OPSD combination degrades severely on Qwen3-1.7B (32.0 vs. 46.1) due to unbounded distillation gradients overwhelming the RL signal, SDAR avoids the observed instability and maintains stable gains. Through its adaptive gating mechanism, it ensures stable optimization and consistent gains across all model scales. **Skills Internalization**Beyond overall performance, SDAR successfully internalizes privileged knowledge rather than superficially relying on it at inference (Lu et al., 2026c). While Skill-GRPO shows a massive performance drop when tested without skills (e.g., 60.2 vs. 80.5 on ALFWorld-3B) and even underperforms vanilla GRPO due to harmful distributional dependencies, SDAR requires no external skills during inference. Yet, it surpasses even the skill-augmented Skill-GRPO* in most settings, achieving 84.4 on ALFWorld-3B and a striking 53.9 (vs. 28.1) on ALFWorld-1.7B. These consistent gains confirm that our token-level gated distillation genuinely transfers underlying knowledge into the policy’s parameters. **Strong Generalization**SDAR also exhibits stronger generalization compared to hybrid baselines such as Skill-SD and RLSD. On Qwen2.5-3B, it outperforms both methods on ALFWorld (84.4 vs. 73.4 for Skill-SD and 79.7 for RLSD) and WebShop. This advantage is most pronounced on the challenging Qwen3-1.7B model, where smaller models may

struggle to utilize retrieved skills effectively. In this regime, Skill-GRPO drops to 21.1% on ALFWorld, well below GRPO’s 46.1%, and RLSO reaches 42.2%. In contrast, SDAR achieves the highest score of 53.9%. By attenuating uncertain negative teacher guidance while preserving positive teacher endorsements, our gating mechanism provides a more robust way to incorporate privileged knowledge without sacrificing generalization.

3.2 Training Dynamics

To elucidate the adaptive behavior of SDAR throughout RL optimization, we monitor two key metrics for the Qwen2.5-7B backbone on ALFWorld in Figure 5. (a) shows that the mean Teacher-Student log-probability gap ($\Delta = E_t[\Delta_t]$) remains consistently negative, indicating that the privileged teacher assigns lower probability than the student to sampled tokens on average. This reveals partial asymmetric trust in privileged guidance regime where naive distillation would actively degrade performance. Crucially, Δ steadily converges toward zero, confirming that the gating mechanism successfully identifies and up-weights the specific subset of tokens where the teacher does provide beneficial signals. To further validate this adaptive filtering, (b) tracks the gate activation ratio (the fraction of tokens where $gt > 0.5$). For the majority of early training, this ratio remains strictly below 0.5, correctly

7
0 20 40 60 80 100 120 140

Step

0.12

0.10

0.08

0.06

0.04

0.02

Teacher Gap Mean

0 20 40 60 80 100 120 140

Step

0.25

0.30

0.35

0.40

0.45

0.50

0.55 Gate Active Ratio

Figure 5: Training Dynamics. Average teacher-student gap (Left) and gate activation ratio (Right) during the training of Qwen2.5-7B-Instruct on ALFWorld.

Table 2: Robust Testing of different skill retrieval methods.

Method	ALFWorld	WebShop-Score	WebShop-Acc
UCB	86.8	+5.6	87.5+6.6
KM	85.9	+4.7	89.4+8.5
Full	83.2	+2.0	87.2+6.3
Random	83.1	+1.9	82.5+1.6
w/o OPSD	81.2	80.9	72.6

suppressing tokens that carry negative signals. However, as the student’s policy evolves, the ratio gradually increases, reflecting that more tokens enter a regime of constructive teacher guidance.

3.3 Robust Analysis

To address the practical concern of whether SDAR heavily relies on high-quality skill retrieval, we fix our optimal configuration ($\lambda = 0.01$, $\beta = 5.0$) and evaluate performance across four retrieval quality tiers (Table 2). All four strategies consistently outperform the pure GRPO baseline (w/o OPSD). Even Random Retrieval—which selects skills with zero task awareness—yields gains of +1.9/+1.6/+1.0 on ALFWorld/WebShop-Score/WebShop-Acc. Higher-quality retrieval further amplifies these benefits: Keyword Matching achieves gains of +4.7/+8.5/+10.2 and even surpasses UCB on WebShop.

These results echo our observation on asymmetric privileged guidance. Low-quality retrieval can introduce mismatched or unstable teacher signals, especially negative guidance from irrelevant skills. Rather than uniformly following such signals, SDAR uses token-level gating to retain positive teacher endorsements while softly attenuating uncertain negative rejections. Thus, the performance gains remain robust across retrieval qualities, suggesting that the uplift stems primarily from gated distillation rather than retrieval fidelity alone.

3.4 Ablation Studies

Token-Level Gating Strategy. As shown in Figure 6, Teacher-Student Gap gating consistently outperforms both the entropy and soft-OR gating strategies (introduced in Section 2.3), achieving a higher asymptotic success rate (~ 0.84) and a steeper performance climb after the initial 100 steps. We attribute this superiority to the directness of the Teacher-Student gap (Δ_t) as an importance signal, which precisely measures the teacher’s disagreement with the student’s chosen token. In contrast, entropy (h_t) acts as an indirect proxy that may erroneously activate on uncertain but already well-handled tokens, while soft-OR dilutes

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0 25 50 75 100 125 150

Step

0.2

0.3
 0.4
 0.5
 0.6
 0.7
 0.8Success Rate
 Entropy
 SOFT-OR Score
 Teacher-Student Gap
 Figure 6: Ablations of Token-level Gating on
 Qwen2.5-3B-Instruct.
 0 25 50 75 100 125 150

Step
 0.2
 0.3
 0.4
 0.5
 0.6
 0.7
 0.8Success Rate
 = 5.0 (Ours)
 = 1.0
 = 10.0
 = 0.0 (No Gate)

Figure 7: Ablations of β on Qwen2.5-3B-Instruct.
 0 25 50 75 100 125 150

Step
 0.0
 0.1
 0.2
 0.3
 0.4
 0.5
 0.6
 0.7
 0.8Success Rate
 = 0.01 (Ours)
 = 0.001
 = 0.1

Figure 8: Ablations of λ on Qwen2.5-3B-Instruct.
 0 25 50 75 100 125 150

Step
 0.2
 0.3
 0.4
 0.5
 0.6
 0.7
 0.8Success Rate
 Reverse KL (Ours)
 Forward KL
 JSD

Figure 9: Ablations of LSDAR type on
 Qwen2.5-7B-Instruct.

the gating signal by triggering when only one score is moderately large, thereby reducing its selectivity. All remaining experiments default to gap gating.

Sharpness β . Figure 7 evaluates the impact of sigmoid sharpness across $\beta \in \{0, 1, 5, 10\}$, where $\beta = 0$ denotes the complete removal of the gating mechanism (i.e., uniform distillation). The optimal performance is achieved at $\beta = 5$, which effectively balances two distinct failure modes: an excessively small β (including the no-gate baseline) applies distillation indiscriminately, thereby inheriting the multi-turn instability of naive OPSP; conversely, an overly large β strictly binarizes the gate, stripping away the smooth modulation necessary for assigning partial credit on borderline tokens.

Distillation Coefficient λ . Figure 8 sweeps the distillation weight $\lambda_{\text{SDAR}} \in \{0.001, 0.01, 0.1\}$, revealing that $\lambda = 0.01$ provides an optimal, steady complementary signal without interfering with the primary RL objective. When λ is increased to 0.1, the distillation gradient overwhelmingly dominates the policy update; since the teacher is on average no confident than the student in multi-turn settings (as evidenced by the negative gap in Figure 5), this over-weighted term forces the student toward inferior behaviors, causing a severe performance decline that overshadows the GRPO reward signal. Conversely, $\lambda = 0.001$ exerts insufficient corrective pressure to meaningfully aid the RL process,

confirming the necessity of a carefully calibrated, moderate coefficient. Distillation Objective. Figure 9 compares three token-level matching objectives on Qwen2.5-7B: reverse KL (our default), forward KL, and Jensen-Shannon divergence (JSD), where JSD is defined as the symmetrized average with respect to the mixture

$$JSD = \frac{1}{2} D_{KL}(\pi_\theta(\cdot | s_t) \parallel \pi_T(\cdot | s_t)) + \frac{1}{2} D_{KL}(\pi_T(\cdot | s_t) \parallel \pi_\theta(\cdot | s_t))$$

Reverse KL clearly outperforms both alternatives, aligning perfectly with our design rationale in Section 2.2: the reverse direction $D_{KL}(\pi_\theta \parallel \pi_T)$ is inherently mode-seeking (Murphy, 2012), encouraging the student to concentrate probability mass only on modes supported by the teacher. In our partial “weak” teacher signals—where the teacher is frequently lost—this selectivity is paramount, as reverse KL naturally down-weights tokens with low teacher probability, thereby seamlessly complementing the explicit gating mechanism. In contrast, the mode-covering nature of forward KL forces the student to spread mass across all teacher-supported tokens, indiscriminately incorporating unreliable guidance, while JSD acts as a symmetric compromise that inherits this detrimental mode-covering tendency, ultimately yielding intermediate performance.

4 Related Work

4.1 Agentic RL

Recent advances in reinforcement learning for LLMs have demonstrated strong effectiveness on verifiable reasoning tasks (Shao et al., 2024; Yu et al., 2025; Guo et al., 2025; Yao et al., 2026; Chen et al., 2026). Building on this progress, LLMs are increasingly extended from static reasoning problems to autonomous agents that operate in dynamic, open-world environments, including GUI automation (Ye et al., 2025), gameplay (Shridhar et al., 2020), and embodied control (Wang et al., 2023). In these settings, agents must make sequential decisions based on environment observations and feedback, making agentic RL a crucial post-training recipe for improving their decision-making capabilities (Lu et al., 2025; Dong et al., 2025; Feng et al., 2025; Lu et al., 2026a;b; Shi et al., 2026).

4.2 OPSD

On-policy distillation (OPD) supervises a student on its own generated sequences, avoiding offline distribution mismatch (Agarwal et al., 2024; Gu et al., 2026). GKD-style methods (Agarwal et al., 2024; Wen et al., 2023) minimize token-level divergences but require full-vocabulary teacher distributions, while PG-style methods (Yang et al., 2026a; Xu et al., 2026) convert discrepancy into token-level rewards but risk high-variance updates. For multi-turn agents, TCO (Wang et al., 2026b) applies a turn-level curriculum to mitigate compounding drift, but relies on rigid schedules. On-Policy Self-Distillation (OPSD) (Zhao et al., 2026; He et al., 2026) further removes the need for a separate teacher by conditioning only on privileged context.

Hybrid Methods Recent works have explored combining RL with distillation to leverage their complementary strengths (Wang et al., 2026a; Yang et al., 2026a; Ding, 2026), but suffer from rigid hand-crafted scheduling or substantially unstable updates. In contrast, our method treats distillation as a strictly separate auxiliary objective with adaptive, bounded, token-level gating, preserving the unbiasedness of the RL advantage while selectively injecting only beneficial teacher signals.

5 Conclusion

We presented SDAR, which reconciles RL and OPSD for multi-turn agent training through a sigmoid gate that lets each token autonomously regulate its distillation intensity. This preserves RL as the unbiased optimization backbone while selectively extracting beneficial teacher signals. Experiments across three benchmarks and three model scales confirm consistent gains over both pure RL and hybrid baselines.

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A Theoretical Analysis

A.1 Design Rationale

The central design question is how the divergence signal should enter optimization. We adopt the reverse-KL-aligned gap

$$\Delta t = \log \pi T(y_t | s_t) - \log \pi \theta(y_t | s_t)$$

rather than forward KL, because it naturally evaluates on student-sampled tokens and avoids the computationally expensive full-vocabulary matching. However, using this raw gap directly as a coefficient would create overly strong, unbounded token-level gradients

during early training or under severe teacher-student mismatch. To resolve this, we wrap the gap in a sigmoid function

$$gt = \sigma(\beta \Delta t),$$

which transforms the raw discrepancy into a bounded and monotone importance weight

$$gt \in (0, 1), \quad \partial gt$$

$$\partial \Delta t$$

$$> 0.$$

This preserves the ordering of token importance while strictly preventing gradient explosion. Finally, we apply a stop-gradient operator to the gate. Detaching gt ensures it acts purely as a confidence weight rather than creating an additional, self-referential optimization pathway, yielding a stable, first-order weighted likelihood update.

A.2 Theoretical Properties

We formalize the stability and curriculum properties of SDAR through the following propositions.

Proposition 1 (Equivalent Weighted Likelihood Form). Assume that both $\log \pi T(y_t | s + t)$ and

gt are detached from gradient computation. Minimizing LSDAR is equivalent, up to an additive constant, to maximizing a token-weighted log-likelihood objective on student-sampled tokens:

$$\text{LSDAR} = C - \text{Agg}(gt \log \pi \theta(y_t | s + t)),$$

where

$$C = \text{Agg}$$

■

$$gt \log \pi T(y_t | s + t)$$

■

is constant with respect to θ .

Proof. By definition,

$$\text{LSDAR} = \text{Agg}$$

■

$$gt(\log \pi T(y_t | s + t) - \log \pi \theta(y_t | s + t))$$

■

$$= \text{Agg}$$

■

$$gt \log \pi T(y_t | s + t)$$

■

$$- \text{Agg}(gt \log \pi \theta(y_t | s + t)) \\ = C - \text{Agg}(gt \log \pi \theta(y_t | s + t)),$$

where the first term $C = \text{Agg}$

■

$$gt \log \pi T(y_t | s + t)$$

■

is constant w.r.t. θ since both gt and $\log \pi T(y_t | s + t)$ are detached.

Proposition 2 (Gradient Form). Under the same assumptions, the gradient of LSDAR is strictly modulated by the bounded scalar gate:

$$\nabla \theta \text{LSDAR} = -\text{Agg}(gt \nabla \theta \log \pi \theta(y_t | s + t)).$$

Proof. From Proposition 1,

$$\nabla \theta \text{LSDAR} = \nabla \theta [C - \text{Agg}(gt \log \pi \theta(y_t | s + t))] \\ = -\text{Agg}(gt \nabla \theta \log \pi \theta(y_t | s + t)) = -\text{Agg}(gt \nabla \theta \log \pi \theta(y_t | s + t)).$$

16

Proposition 3 (Monotonicity and Smoothness of the Gate). The gate $gt = \sigma(\beta \Delta t)$ is strictly increasing in Δt , inducing an online token-level curriculum where larger discrepancies receive stronger weights. Its derivative satisfies

$$\partial gt$$

$$\partial \Delta t$$

$$= \beta \sigma(\beta \Delta t)(1 - \sigma(\beta \Delta t)) \in (0, \beta/4].$$

Proof. By the chain rule,

$$\partial gt$$

$$\partial \Delta t$$

$$= \beta \sigma'(\beta \Delta t).$$

Since the logistic sigmoid satisfies

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)) > 0 \quad \forall z \in \mathbb{R},$$

we obtain

$$\partial gt$$

$$\partial \Delta t$$

$$= \beta \sigma(\beta \Delta t)(1 - \sigma(\beta \Delta t)) > 0.$$

$$\text{Let } u = \sigma(\beta \Delta t) \in (0, 1):$$

$$u(1-u) \leq$$

$$\text{■ } u + (1-u)$$

2
 $\mathbb{E}[\frac{\partial g_t}{\partial \Delta t}] = 1$
4
 $\frac{\partial g_t}{\partial \Delta t} = \beta u(1-u) \leq \beta$
4 .

Proposition 4(Bounded Auxiliary Gradient). Assume that $\|\nabla_{\theta} \log \pi_{\theta}(y_t | s_t)\| \leq B$ for each valid token. Then the gate cannot amplify the auxiliary gradient beyond the unweighted likelihood gradient:

$\|\nabla_{\theta} \text{LSAR}\| \leq \text{Agg}(B \cdot t)$.

Proof. By Proposition 2,
 $\|\nabla_{\theta} \text{LSAR}\| = \|\text{Agg}(g_t \nabla_{\theta} \log \pi_{\theta}(y_t | s_t))\|$
 $\leq \text{Agg}(g_t \|\nabla_{\theta} \log \pi_{\theta}(y_t | s_t)\|)$
 $\leq \text{Agg}(1 \cdot B \cdot t) = \text{Agg}(B \cdot t)$,
where the first inequality is the triangle inequality and the second uses $0 < g_t < 1$ and $\|\nabla_{\theta} \log \pi_{\theta}(y_t | s_t)\| \leq B$.

Proposition 5(Effect of Not Detaching the Gate). Without stop-gradient on the gate, the non-detached token loss $\tilde{\ell}_t = \sigma(\beta \Delta t) \Delta t$ introduces an unstable self-referential coupling term into the gradient:

$\nabla_{\theta} \tilde{\ell}_t = -$
 $\mathbb{E}[\frac{\partial g_t}{\partial \Delta t} + \beta \Delta t \cdot g_t(1-g_t)]$
 $\mathbb{E}[\nabla_{\theta} \log \pi_{\theta}(y_t | s_t)]$.

Proof. Write $\tilde{\ell}_t = g_t \Delta t$. Since $\log \pi_{\theta}(y_t | s_t)$ is constant w.r.t. θ ,
 $\nabla_{\theta} \Delta t = \nabla_{\theta} [\log \pi_{\theta}(y_t | s_t) - \log \pi_{\theta}(y_t | s_t)] = -\nabla_{\theta} \log \pi_{\theta}(y_t | s_t)$.
By the chain rule $\frac{\partial g_t}{\partial \Delta t} = \sigma'(\beta \Delta t)$,
 $\nabla_{\theta} g_t = \beta \sigma'(\beta \Delta t) \nabla_{\theta} \Delta t = \beta g_t(1-g_t) \nabla_{\theta} \Delta t = -\beta g_t(1-g_t) \nabla_{\theta} \log \pi_{\theta}(y_t | s_t)$.
Applying the product rule,
 $\nabla_{\theta} \tilde{\ell}_t = (\nabla_{\theta} g_t) \Delta t + g_t (\nabla_{\theta} \Delta t)$
 $= [-\beta g_t(1-g_t) \nabla_{\theta} \log \pi_{\theta}(y_t | s_t)] \Delta t + g_t [-\nabla_{\theta} \log \pi_{\theta}(y_t | s_t)]$
 $= -\beta g_t(1-g_t) \Delta t \nabla_{\theta} \log \pi_{\theta}(y_t | s_t) - g_t \nabla_{\theta} \log \pi_{\theta}(y_t | s_t)$
 $= -$
 $\mathbb{E}[\frac{\partial g_t}{\partial \Delta t} + \beta \Delta t \cdot g_t(1-g_t)]$
 $\mathbb{E}[\nabla_{\theta} \log \pi_{\theta}(y_t | s_t)]$.

17

B Algorithm

The full procedure of SDAR is presented in Algorithm 1. We compare against five baselines listed below:

- GRPO(Shao et al., 2024) (Algorithm 2): RL baseline that optimizes the policy via a clipped surrogate objective with group-relative advantages.
- OPSD(Zhao et al., 2026) (Algorithm 3): an on-policy self-distillation method that distills token-level knowledge from a frozen reference policy π_{ref} into the student.
- Skill-SD(Wang et al., 2026a) (Algorithm 4): a hybrid method that augments GRPO with an importance-weighted K3-divergence distillation loss, using retrieved skills as privileged context to construct the teacher signal.
- GRPO+OPSD(Algorithm 5): a hybrid method that simply adds the OPSD distillation loss from π_{ref} as an auxiliary objective on top of GRPO training.
- RLSD(Yang et al., 2026a) (Algorithm 6): a hybrid method that re-weights GRPO's advantages with self-teacher's gap.

Algorithm 1SDAR

Require: Policy π_{θ} , task set S , skill library $E=\{e_1, \dots, e_M\}$, group size G , mixing coefficient λ , sharpness β , clip bound κ

- 1:foreach training iteration do
- 2:Sample a batch of tasks $\{x\}$ from S
- 3:foreach task x do
- 4:Retrieve skills $c + \text{from } E \sim \text{UCB} / \text{KM} / \text{Full} / \text{Random}$
- 5:// Step 1: On-policy rollout
- 6:Sample Responses $\{y(1), \dots, y(G)\} \sim \pi_{\theta}(\cdot | x)$
- 7:// Step 2: Sequence-level advantage from environment
- 8:for $i=1, \dots, G$ do
- 9:Obtain reward $R(x, y(i))$ from environment interaction
- 10:end for
- 11:Compute $A(i) = R(x, y(i)) - \mu_G$
- 12:// Step 3: GRPO policy loss

$\mathbb{E}[\text{Group-relative advantage}]$

```

13:for i=1, . . . ,Gdo
14:fort=1, . . . ,|y (i) |do
15:r (i)
t  $\leftarrow \pi \theta(y(i))$ 
t |s (i)
t )/ $\pi \theta_{old}(y(i))$ 
t |s (i)
t )
16:end for
17:end for
18:Compute L GRPO via clipped surrogate with {A (i), r (i)}
t }
19:// Step 4: Token-level gated distillation
20:for i=1, . . . ,Gdo
21:Compute teacher logits via forward pass with (x, c +, y (i) )
22:fort=1, . . . ,|y (i) |do
23: $\Delta t \leftarrow \text{sg}(\log \pi \theta(y(i))$ 
t |s +
t )- $\log \pi \theta(y(i))$ 
t |s t))
24:g t  $\leftarrow \sigma(\beta \cdot \Delta t)$ 
25: $\blacksquare t \leftarrow g t \cdot (\log \pi \theta(y(i))$ 
t |s +
t )- $\log \pi \theta(y(i))$ 
t |s t))
26:end for
27:end for
28:L SDAR  $\leftarrow 1$ 
G  $\sum G$ 
i=1 Agg( $\blacksquare(i)$ 
t )
29:// Step 5: Joint policy update
30:Update  $\theta$  by minimizing  $L(\theta) = L \text{ GRPO}(\theta) + \lambda \cdot L \text{ SDAR}(\theta)$ 
31:end for
32:end for
18
Algorithm 2 GRPO
Require: Policy  $\pi \theta$ , task set S, group size G, clip bounds  $\blacksquare_{lo}, \blacksquare_{hi}$ , dual-clip constant c
1:foreach training iteration do
2:Sample a batch of tasks {x} from S
3:foreach task x do
4:// Step 1: On-policy rollout
5:Sample G responses {y (1), . . . , y (G) }  $\sim \pi \theta(\cdot |x)$ 
6:// Step 2: Sequence-level advantage from environment
7:for i=1, . . . ,Gdo
8:Obtain reward R(x, y (i) ) from environment interaction
9:end for
10:Compute A (i) =  $R(x, y (i) ) - \mu_G$ 
 $\sigma G$ 
 $\blacksquare$  Group-relative advantage
11:// Step 3: Clipped surrogate policy loss
12:for i=1, . . . ,Gdo
13:fort=1, . . . ,|y (i) |do
14:r (i)
t  $\leftarrow \pi \theta(y(i))$ 
t |s (i)
t )/ $\pi \theta_{old}(y(i))$ 
t |s (i)
t )
15:L 1  $\leftarrow -A (i)r(i)$ 
t
16:L 2  $\leftarrow -A (i) \text{clip}(r(i)$ 
t ,  $1-\blacksquare_{lo}, 1+\blacksquare_{hi})$ 
17: $\blacksquare (i)$ 
t  $\leftarrow$ 
(
min( $-A (i)c, \max(L 1, L 2)$ ) if A (i) < 0
max(L 1, L 2) otherwise
18:end for
19:end for
20:L GRPO  $\leftarrow \text{Agg}(\{\blacksquare(i)$ 
t })
21:// Step 4: Policy update

```

```

22:Update $\theta$  by minimizing  $L(\theta) = L_{GRPO}(\theta)$ 
23:end for
24:end for
Algorithm 3 OPSD
Require: Policy  $\pi_\theta$ , frozen reference  $\pi_{ref}$ , task set  $S$ , group size  $G$ , KL coefficient  $\alpha$ 
1:foreach training iteration do
2:Sample a batch of tasks  $\{x\}$  from  $S$ 
3:foreach task  $x$  do
4:// Step 1: On-policy rollout
5:Sample  $G$  responses  $\{y(1), \dots, y(G)\} \sim \pi_\theta(\cdot | x)$ 
6:// Step 2: Token-level KL distillation from reference
7:for  $i=1, \dots, G$  do
8:for  $t=1, \dots, |y(i)|$  do
9: $d(i)$ 
 $t \leftarrow \log \pi_\theta(y(i))$ 
 $t \leftarrow |s_t| - \log \pi_{ref}(y(i))$ 
 $t \leftarrow |s_t| \cdot D_{KL}(\pi_\theta || \pi_{ref})$ 
10:end for
11:end for
12: $L_{OPSD} \leftarrow \alpha \cdot \text{Agg}(\{d(i)\})$ 
13:// Step 3: Policy update
14:Update  $\theta$  by minimizing  $L(\theta) = L_{OPSD}(\theta)$ 
15:end for
16:end for
19
Algorithm 4 Skill-SD
Require: Policy  $\pi_\theta$ , task set  $S$ , skill library  $E=\{e_1, \dots, e_M\}$ , group size  $G$ , distillation coefficient  $\lambda$ , clip bound  $\beta$ 
1:foreach training iteration do
2:Sample a batch of tasks  $\{x\}$  from  $S$ 
3:foreach task  $x$  do
4:Retrieve skill  $c +$  from  $E$  via UCB
5:// Step 1: On-policy rollout
6:Sample  $G$  responses  $\{y(1), \dots, y(G)\} \sim \pi_\theta(\cdot | x)$ 
7:// Step 2: Sequence-level advantage from environment
8:for  $i=1, \dots, G$  do
9:Obtain reward  $R(x, y(i))$  from environment interaction
10:end for
11:Compute  $A(i) = R(x, y(i)) - \mu_G$ 
 $\sigma_G$ 
12:// Step 3: GRPO policy loss (same as Algorithm 2)
13:Compute  $L_{GRPO}$  via clipped surrogate with  $\{A(i), r(i)\}$ 
14:// Step 4: Importance-weighted K3 distillation
15:for  $i=1, \dots, G$  do
16:Compute teacher log-probs via forward pass with  $(x, c +, y(i))$ 
17:for  $t=1, \dots, |y(i)|$  do
18: $d_t \leftarrow \log \pi_\theta(y(i))$ 
 $t \leftarrow |s_t| - \log \pi_\theta(y(i))$ 
 $t \leftarrow |s_t| \cdot \text{Student-Teacher}$ 
19: $k_t \leftarrow \exp(-d_t) - 1 + d_t$   $\blacksquare$  K3 divergence
20: $p_t \leftarrow \exp(\log \pi_\theta(y(i))$ 
 $t \leftarrow |s_t| - \log \pi_{old}(y(i))$ 
 $t \leftarrow |s_t| \cdot \text{On-policy IS ratio}$ 
21: $\blacksquare(i)$ 
 $t \leftarrow p_t \cdot k_t$ 
22:end for
23:end for
24: $L_{Skill-SD} \leftarrow \text{Agg}(\{\blacksquare(i)\})$ 
25:// Step 5: Joint policy update
26:Update  $\theta$  by minimizing  $L(\theta) = L_{GRPO}(\theta) + \lambda \cdot L_{Skill-SD}(\theta)$ 
27:end for
28:end for
20
Algorithm 5 GRPO+OPSD
Require: Policy  $\pi_\theta$ , frozen reference  $\pi_{ref}$ , task set  $S$ , group size  $G$ , KL coefficient  $\alpha$ , clip bound  $\beta$ 
1:foreach training iteration do
2:Sample a batch of tasks  $\{x\}$  from  $S$ 

```



```

3:foreach taskxdo
4:// Step 1: On-policy rollout
5:SampleGresponses{y (1), . . . ,y(G) }  $\sim \pi_{\theta}(\cdot |x)$ 
6:// Step 2: Sequence-level advantage from environment
7:fori=1, . . . ,Gdo
8:Obtain rewardR(x,y (i) )from environment interaction
9:end for
10:ComputeA (i) = R(x,y (i) )- $\mu$  G
 $\sigma$ G
■Group-relative advantage
11:// Step 3: GRPO policy loss (same as Algorithm 2)
12:ComputeL GRPO via clipped surrogate with{A (i),r (i)
t }
13:// Step 4: Token-level KL penalty toward $\pi$  ref
14:fori=1, . . . ,Gdo
15:fort=1, . . . ,|y (i) |do
16:d (i)
t  $\leftarrow \log \pi_{\theta}(y(i))$ 
t |s t)- $\log \pi_{\text{ref}}(y(i))$ 
t |s t)
17:end for
18:end for
19:L OPSD  $\leftarrow \alpha \cdot \text{Agg}(\{d(i)
t \})$ 
20:// Step 5: Joint policy update
21:Update $\theta$ by minimizingL( $\theta$ ) =L GRPO( $\theta$ ) +L OPSD( $\theta$ )
22:end for
23:end for
21
Algorithm 6RLSD
Require: Policy  $\pi_{\theta}$ , task set S, skill library E={e 1, . . . , eM}, group size G, mixing coefficient $\lambda$ , weight clip bound■ w, policy clip bound■
1:foreach training iterationdo
2:Sample a batch of tasks{x}fromS
3:foreach taskxdo
4:Retrieve skillc + fromE■UCB / KM / Full / Random
5:// Step 1: On-policy rollout
6:SampleGresponses{y (1), . . . ,y(G) }  $\sim \pi_{\theta}(\cdot |x)$ 
7:// Step 2: Sequence-level advantage from environment
8:fori=1, . . . ,Gdo
9:Obtain rewardR(x,y (i) )from environment interaction
10:end for
11:ComputeA (i) = R(x,y (i) )- $\mu$  G
 $\sigma$ G
■Group-relative advantage
12:// Step 3: Token-level advantage reweighting via teacher
13:fori=1, . . . ,Gdo
14:Compute teacher log-probs via forward pass with(x,c +,y (i) )
15:fort=1, . . . ,|y (i) |do
16: $\delta$  t  $\leftarrow \log \pi_{\theta}(y(i))$ 
t |s +
t )- $\log \pi_{\theta}(\text{old}(y(i))$ 
t |s t)■Teacher-Student gap
17:w t  $\leftarrow \text{clip}(\exp(\text{sign}(A(i)) \cdot \delta t), 1-\text{■} w, 1+\text{■} w)$ 
18:  $\hat{A}(i)$ 
t  $\leftarrow A(i) \cdot [(1-\lambda) + \lambda \cdot w t]$ 
19:end for
20:end for
21:// Step 4: Clipped surrogate with token-level advantages
22:ComputeL RLSD via clipped surrogate with{  $\hat{A}(i)$ 
t ,r (i)
t }
23:// Step 5: Policy update
24:Update $\theta$ by minimizingL( $\theta$ ) =L RLSD( $\theta$ )
25:end for
26:end for
22
C Hyperparameters
Table 3 summarizes the method-specific hyperparameters used for all baselines and SDAR across our experiments.
Table 3:Hyperparameters.  $\eta$ : learning rate; G: group size; ■: PPO clip ratio;  $\lambda$ : distillation loss coefficient;  $\beta$ : sigmoid gate sharpness;  $\alpha$ KL: KL penalty coefficient toward the reference policy; SRS: skill retrieval strategy (KM = keyword matching).

```

Method η λ β α KL SRS
 GRPO 10 -6 8 0.2 - - 0.01 -
 Skill-GRPO 10 -6 8 0.2 - - 0.01 KM
 OPSD 10 -6 - - 0.01 5.0 0.01 KM
 Skill-SD 10 -6 8 0.2 0.001 - 0.01 KM
 GRPO+OPSD 10 -6 8 0.2 0.01 0.0 0.01 KM
 RLSD 10 -6 8 0.2 0.5 - 0.01 KM
 SDAR (Ours) 10 -6 8 0.2 0.01 5.0 0.01 KM

D Training Dynamics

We present the full training dynamics of SDAR across all model scales and environments in Figures 10-14, tracking five diagnostic metrics throughout training.

0 20 40 60 80 100 120 140

0.30

0.35

0.40

0.45

0.50ALFWorld

Qwen2.5-3B

0 20 40 60 80 100 120 140

0.25

0.30

0.35

0.40

0.45

0.50

0.55

Qwen2.5-7B

0 20 40 60 80 100 120 140

0.20

0.22

0.24

0.26

0.28

0.30

0.32

0.34

Qwen3-1.7B

0 20 40 60 80 100 120 140

0.25

0.30

0.35

0.40

0.45WebShop

0 20 40 60 80 100 120 140

0.25

0.30

0.35

0.40

0.45

0.50

0 20 40 60 80 100 120 140

0.20

0.25

0.30

0.35

0.40

0 20 40 60 80 100 120 140

Step

0.175

0.200

0.225

0.250

0.275

0.300

0.325

0.350

0.375Search

0 20 40 60 80 100 120 140

Step

0.25

0.30

0.35

0.40

0 20 40 60 80 100 120 140
Step
0.125
0.150
0.175
0.200
0.225
0.250
0.275
0.300
0.325

Figure 10: Gate Active Ratio when training Qwen2.5-3B, Qwen2.5-7B and Qwen3-1.7B on ALFWorld, WebShop and Search-QA.

23
0 20 40 60 80 100 120 140
0.43
0.44
0.45
0.46
0.47
ALFWorld
Qwen2.5-3B
0 20 40 60 80 100 120 140
0.45
0.46
0.47
0.48
0.49
Qwen2.5-7B
0 20 40 60 80 100 120 140
0.43
0.44
0.45
0.46
0.47
Qwen3-1.7B
0 20 40 60 80 100 120 140
0.440
0.445
0.450
0.455
0.460
0.465
0.470
0.475
WebShop
0 20 40 60 80 100 120 140
0.45
0.46
0.47
0.48
0.49
0 20 40 60 80 100 120 140
0.43
0.44
0.45
0.46
0.47
0.48
0 20 40 60 80 100 120 140
Step
0.450
0.455
0.460
0.465
0.470
0.475
0.480
0.485
Search
0 20 40 60 80 100 120 140
Step
0.460
0.465
0.470
0.475

0.480
0.485
0.490
0 20 40 60 80 100 120 140
Step
0.450
0.455
0.460
0.465
0.470
0.475
0.480

Figure 11: Gate Mean when training Qwen2.5-3B, Qwen2.5-7B and Qwen3-1.7B on ALF-World, WebShop and Search-QA.

E Prompt

Figures 15-17 present the full prompt templates used by SDAR for the three evaluation environments, where {skill context} is populated with the retrieved skill during training and left empty at inference time.

24
0 20 40 60 80 100 120 140
0.035
0.040
0.045
0.050
0.055
0.060
0.065
0.070 ALFWorld
Qwen2.5-3B
0 20 40 60 80 100 120 140
0.02
0.03
0.04
0.05
0.06
Qwen2.5-7B
0 20 40 60 80 100 120 140
0.025
0.030
0.035
0.040
0.045
0.050
Qwen3-1.7B
0 20 40 60 80 100 120 140
0.025
0.030
0.035
0.040
0.045
0.050
0.055
0.060 WebShop
0 20 40 60 80 100 120 140
0.02
0.03
0.04
0.05
0.06
0 20 40 60 80 100 120 140
0.015
0.020
0.025
0.030
0.035
0.040
0.045
0.050
0 20 40 60 80 100 120 140
Step
0.015
0.020
0.025

```

0.030
0.035
0.040
0.045Search
0 20 40 60 80 100 120 140
Step
0.015
0.020
0.025
0.030
0.035
0.040
0.045
0 20 40 60 80 100 120 140
Step
0.010
0.015
0.020
0.025
0.030
0.035
0.040
0.045
Figure 12:OPSD Losswhen training Qwen2.5-3B, Qwen2.5-7B and Qwen3-1.7B on ALF-
World, WebShop and Search-QA.
25
0 20 40 60 80 100 120 140
0.18
0.16
0.14
0.12
0.10
0.08
ALFWorld
Qwen2.5-3B
0 20 40 60 80 100 120 140
0.12
0.10
0.08
0.06
0.04
0.02
Qwen2.5-7B
0 20 40 60 80 100 120 140
0.5
0.4
0.3
0.2
Qwen3-1.7B
0 20 40 60 80 100 120 140
0.12
0.10
0.08
0.06
0.04
WebShop
0 20 40 60 80 100 120 140
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0 20 40 60 80 100 120 140
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0 20 40 60 80 100 120 140

```

Step
0.12
0.10
0.08
0.06
0.04

Search
0 20 40 60 80 100 120 140

Step
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0 20 40 60 80 100 120 140

Step
0.40
0.35
0.30
0.25
0.20
0.15
0.10

Figure 13: Teacher-Student Gap when training Qwen2.5-3B, Qwen2.5-7B and Qwen3-1.7B on ALFWorld, WebShop and Search-QA.

26
0 20 40 60 80 100 120 140

1
2
3
4
5

6ALFWorld

Qwen2.5-3B

0 20 40 60 80 100 120 140

1
2
3
4
5
6
7
8

Qwen2.5-7B

0 20 40 60 80 100 120 140

0.0
0.5
1.0
1.5
2.0
2.5

Qwen3-1.7B

0 20 40 60 80 100 120 140

0
1
2
3
4
5
6

7WebShop

0 20 40 60 80 100 120 140

0
2
4
6
8

0 20 40 60 80 100 120 140

0
1
2

3
4
5
6
0 20 40 60 80 100 120 140
Step
0.25
0.30
0.35
0.40
0.45
0.50
Search
0 20 40 60 80 100 120 140
Step
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0 20 40 60 80 100 120 140
Step
0.25
0.30
0.35
0.40
0.45
0.50

Figure 14: Reward Curve when training Qwen2.5-3B, Qwen2.5-7B and Qwen3-1.7B on ALFWorld, WebShop and Search-QA.

Prompt of SDAR on ALFWorld

You are an expert agent operating in the ALFRED Embodied Environment. Your task is to: {task description}.

{skill context}

Prior to this step, you have already taken {step count} step(s). Below are the most recent {history length} observations and the corresponding actions you took: {action history}

You are now at step {current step} and your current observation is:

{current observation}

Your admissible actions of the current situation are: [{admissible actions}].

Now it's your turn to take an action. You should first reason step-by-step about the current situation. This reasoning process MUST be enclosed within <think> </think> tags. Once you've finished your reasoning, you should choose an admissible action for current step and present it within <action> </action> tags.

Figure 15: Prompt template used by SDAR for the ALFWorld task environment.

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Prompt of SDAR on Search-based QA

You are an expert agent tasked with answering the given question step-by-step.

{skill context}

Your question: {task description}.

Prior to this step, you have already taken {step count} step(s). Below is the inter-action history where <search> </search> wrapped your past search queries and <information> </information> wrapped the corresponding search results returned by the external search engine. History:

{memory context}

Now it's your turn to respond for the current step. You should first conduct a reasoning process. This process MUST be enclosed within <think> </think> tags. After completing your reasoning, choose only one of the following actions (do not perform both):

1. If you find you lack some knowledge, you MUST call a search engine to get more external information using format: <search> your query </search>.
2. If you have enough knowledge to answer the question confidently, provide your final answer within <answer> </answer> tags, without detailed illustrations. For example, <answer>Beijing</answer>.

Figure 16: Prompt template used by SDAR for the Search-based QA task environment.

Prompt of SDAR on WebShop

You are an expert autonomous agent operating in the WebShop e-commerce environment.

{skill context}

Your task is to: {task description}.

Prior to this step, you have already taken {step count} step(s). Below are the most recent {history length} observations and the corresponding actions you took: {action history}

You are now at step {current step} and your current observation is:

{current observation}.

Your admissible actions of the current situation are: [{available actions}].

Now it's your turn to take one action for the current step. You should first reason step-by-step about the current situation, then think carefully which admissible action best advances the shopping goal. This reasoning process MUST be enclosed within <think> </think> tags. Once you've finished your reasoning, you should choose an admissible action for current step and present it within <action> </action> tags.

Figure 17: Prompt template used by SDAR for the WebShop task environment.